CHAPTER 4 – LINEAR AND QUADRATIC FUNCTIONS

Exercise 1.

In Problems 1–6:

(a) Determine the slope and y-intercept of each linear function.

(b) Find the average rate of change of each function.

(c) Graph each function. Label the intercepts.

(d) Determine whether the function is increasing, decreasing, or constant.

1. $f(x) = 2x - 5$ **2.** $g(x) = -4x + 7$ **3.** $h(x) = \frac{4}{5}x - 6$

4. $F(x) = -\frac{1}{3}x + 1$ **5.** $G(x) = 4$ **6.** $H(x) = -3$

Exercise 2.

In Problems 7 and 8, determine whether the function is linear or nonlinear. If the function is linear, state its slope.

7.

x	$y = f(x)$
−1	−2
0	3
1	8
2	13
3	18

8.

x	$y = g(x)$
−1	−3
0	4
1	7
2	6
3	1

Exercise 3.

In Problems 9–14, graph each quadratic function using transformations (shifting, compressing, stretching, and/or reflecting).

9. $f(x) = (x - 2)^2 + 2$ **10.** $f(x) = (x + 1)^2 - 4$ **11.** $f(x) = -(x - 4)^2$

12. $f(x) = (x - 1)^2 - 3$ **13.** $f(x) = 2(x + 1)^2 + 4$ **14.** $f(x) = -3(x + 2)^2 + 1$

Exercise 4.

In Problems 15–24, (a) graph each quadratic function by determining whether its graph opens up or down and by finding its vertex, axis of symmetry, y-intercept, and x-intercepts, if any. (b) Determine the domain and the range of the function. (c) Determine where the function is increasing and where it is decreasing.

19. $f(x) = -4x^2 + 4x$ **20.** $f(x) = 9x^2 - 6x + 3$ **21.** $f(x) = \frac{9}{2}x^2 + 3x + 1$ **22.** $f(x) = -x^2 + x + \frac{1}{2}$

23. $f(x) = 3x^2 + 4x - 1$ **24.** $f(x) = -2x^2 - x + 4$

Exercise 5.

In Problems 25–30, determine whether the given quadratic function has a maximum value or a minimum value, and then find the value.

25. $f(x) = 3x^2 - 6x + 4$ **26.** $f(x) = 2x^2 + 8x + 5$ **27.** $f(x) = -x^2 + 8x - 4$

28. $f(x) = -x^2 - 10x - 3$ **29.** $f(x) = -3x^2 + 12x + 4$ **30.** $f(x) = -2x^2 + 4$

Exercise 6.

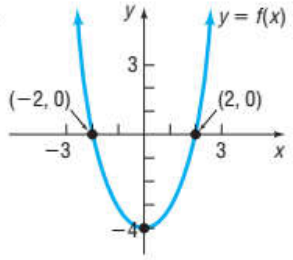
In problems 31-34, solve each quadratic inequality

31. $x^2 + 6x - 16 < 0$ **32.** $3x^2 - 2x - 1 \geq 0$ **33.** $3x^2 \geq 14x + 5$ **34.** $4x^2 < 13x - 3$

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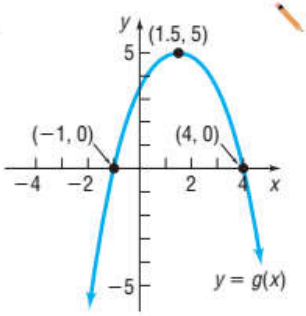
Exercise 7.
In Problems 3–6, use the figure to solve each inequality

3.



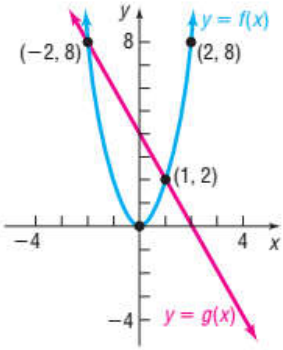
(a) $f(x) > 0$
(b) $f(x) \leq 0$

4.



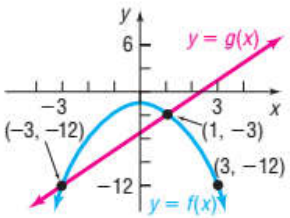
(a) $g(x) < 0$
(b) $g(x) \geq 0$

5.



(a) $g(x) \geq f(x)$
(b) $f(x) > g(x)$

6.



(a) $f(x) < g(x)$
(b) $f(x) \geq g(x)$

Exercise 8.
In Problems 35 and 36, find the quadratic function for which:

35. Vertex is $(-1, 2)$; contains the point $(1, 6)$

36. Vertex is $(3, -4)$; contains the point $(4, 2)$

Exercise 9.
Comparing Phone Companies Marissa must decide between one of two companies as her long-distance phone provider. Company A charges a monthly fee of $7.00 plus $0.06 per minute, while Company B does not have a monthly fee, but charges $0.08 per minute.
(a) Find a linear function that relates cost, C, to total minutes on the phone, x, for each company.
(b) Determine the number of minutes x for which the bill from Company A will equal the bill from Company B.
(c) Over what interval of minutes x will the bill from Company B be less than the bill from Company A?

Exercise 10.
Sales Commissions Bill was just offered a sales position for a computer company. His salary would be $15,000 per year plus 1% of his total annual sales.
(a) Find a linear function that relates Bill's annual salary, S, to his total annual sales, x.
(b) In 2010, Bill had total annual sales of $1,000,000. What was Bill's salary?
(c) What would Bill have to sell to earn $100,000?
(d) Determine the sales required of Bill for his salary to exceed $150,000

Exercise 11.
Demand Equation The price p (in dollars) and the quantity x sold of a certain product obey the demand equation $p = -\frac{1}{10}x + 150 \quad 0 \leq x \leq 1500$
(a) Express the revenue R as a function of x.
(b) What is the revenue if 100 units are sold?
(c) What quantity x maximizes revenue? What is the maximum revenue?
(d) What price should the company charge to maximize revenue?

Exercise 12.
Landscaping A landscape engineer has 200 feet of border to enclose a rectangular pond. What dimensions will result in the largest pond?

HOMEWORKS:
Exercise 1: 2
Exercise 2: 8
Exercise 3: 10
Exercise 4: 22
Exercise 5: 26
Exercise 6: 32
Exercise 7: 4
Exercise 8: 35, 36
Exercise 9
Exercise 10.

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